

Project Orca: Final Report

1. Title

Project Title: Orca: An End-to-End Deep Learning System for Automated Brain Tumor Segmentation and Volumetric Analysis

2. Abstract

The manual segmentation of brain tumors from multi-parametric Magnetic Resonance Imaging (mpMRI) is a cornerstone of neuro-oncological diagnosis and treatment planning. However, this process is notoriously time-consuming, labor-intensive, and subject to significant inter-observer variability, impacting clinical consistency. This report details **Project Orca**, an end-to-end deep learning system designed to overcome these challenges by providing automated, precise, and reproducible brain tumor segmentation.

The core of the system is a **3D Attention U-Net model**, which was rigorously trained and validated on the internationally recognized **BraTS 2023 dataset**. The model processes four standard MRI modalities (T1, T1-contrast enhanced, T2, T2-FLAIR) to produce a multi-class segmentation mask delineating three clinically relevant sub-regions: the Whole Tumor (WT), the Tumor Core (TC), and the Enhancing Tumor (ET).

Upon evaluation, the model achieved a mean Dice Similarity Coefficient of **0.7812** on the validation set. On a blind test set of 150 unseen patient scans, it achieved a mean Dice score of 0.6892 for the Whole Tumor, 0.5947 for the Tumor Core, and 0.5079 for the Enhancing Tumor. These results indicate decent performance in identifying larger tumor structures, while highlighting the known challenge of segmenting smaller, less distinct sub-regions.

The entire pipeline from data ingestion and preprocessing to model inference and post-processing has been encapsulated in an interactive, user-friendly web application built with a Flask backend. This deployed system provides both 2D slice-by-slice inspection and interactive 3D mesh visualizations, offering a powerful tool for quantitative and objective tumor assessment that can enhance the efficiency and reliability of clinical workflows.

3. Introduction

3.1 Clinical Background and Problem Statement

In modern neuro-oncology, multi-parametric MRI is the primary imaging modality for the diagnosis, treatment planning, and monitoring of brain tumors. The accurate delineation of the tumor and its constituent sub-regions such as the necrotic core, the enhancing active tumor, and peritumoral edema is critical for determining tumor grade, planning surgical resection, guiding radiotherapy, and assessing treatment response.

The current clinical standard relies on radiologists manually tracing the boundaries of these regions on hundreds of 2D MRI slices. This process suffers from three fundamental limitations:

1. **It is extremely time-consuming**, often taking over an hour per patient, which creates bottlenecks in clinical workflows.
2. **It is labor-intensive**, requiring the sustained focus of highly trained medical experts.

3. **It suffers from significant inter- and intra-observer variability**, where different experts (or the same expert at different times) produce different segmentations, leading to inconsistencies in clinical assessment and treatment planning.

3.2 Project Orca: An Automated Solution

Project Orca was conceived to directly address these limitations by developing a robust, automated deep learning tool for brain tumor segmentation. The project's goal is to provide a quantitative, objective, and reproducible method to assess tumor morphology, thereby improving the efficiency, accuracy, and reliability of the entire clinical pipeline.

3.3 Project Objectives

The objectives of the project were formally defined as follows:

- **Primary Objective:** To design, build, and train a deep learning model that automatically segments brain tumors into their constituent sub-regions (enhancing tumor, tumor core, whole tumor) from multi-modal MRI scans.
 - **Secondary Objectives:**
 1. **Achieve High Accuracy:** The model's performance is measured using standard segmentation metrics, Dice Similarity Coefficient (DSC).
 2. **Develop a Multi-Modal System:** The model must effectively process and fuse information from the four standard MRI modalities (T1, T1ce, T2, FLAIR) to produce a good segmentation.
 3. **Generate Quantitative Outputs:** The system must output not just a visual mask but also quantitative data, such as the volume and composition percentage of each tumor sub-region.
 4. **Create an End-to-End Solution:** The project aims to extend beyond a simple segmentation mask by integrating the model into a user-friendly application that provides visualization and preliminary reporting capabilities.
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4. Literature Review

The development of Project Orca was built upon a well-established body of research in medical image segmentation.

4.1 Foundational Architectures in Medical Segmentation

- **U-Net:** This architecture is the de-facto standard for biomedical image segmentation. Its encoder-decoder structure with "skip connections" is uniquely effective. The encoder path captures high-level contextual features, while the skip connections allow the decoder path to reuse low-level, fine-grained spatial details, enabling precise localization and boundary delineation.
- **3D U-Net / V-Net:** Brain tumors are inherently 3D structures. 3D extensions of the U-Net process entire MRI volumes at once, rather than slice-by-slice. This allows the model to learn from the crucial inter-slice spatial context, leading to more coherent and anatomically plausible segmentations.
- **Attention U-Net (We Used):** An enhancement to the U-Net that incorporates attention mechanisms. These "attention gates," placed on the skip connections, learn to focus on the most relevant parts of the input image features, suppressing irrelevant background noise. This is particularly valuable for medical imaging, where target structures can be small and surrounded by complex anatomy.

- **Transformer-based Models (e.g., Swin-Unet):** More recently, Vision Transformers (ViTs) have been adapted for segmentation tasks. Models like the Swin-Unet can capture long-range dependencies in an image more effectively than convolutional networks, offering a promising alternative to U-Net-based architectures.

4.2 Benchmarks and Datasets

The **Brain Tumor Segmentation (BraTS) Challenge dataset** is the universally recognized gold standard for this task. It provides thousands of high-quality, multi-modal MRI scans with manually annotated ground truth segmentations from multiple institutions. It is the primary resource for training, validating, and benchmarking models in a fair and reproducible manner. Standard evaluation metrics include the **Dice Similarity Coefficient (DSC)** and **Intersection over Union (IoU)**, which measure overlap, and the **Hausdorff Distance**, which measures boundary distance.

5. Dataset and Methodology

5.1 Dataset: BraTS 2023

This project utilized the **BraTS 2023 dataset**, sourced from official repositories after agreeing to the data usage terms. The dataset is released under the **Creative Commons Attribution–NonCommercial 4.0 International (CC BY-NC 4.0)** license, and all data is de-identified for research use.

5.2 Exploratory Data Analysis (EDA)

An initial EDA revealed that while all 3D scans were consistent in dimension (`240x240x155`) and voxel spacing (`1.0, 1.0, 1.0`), the intensity values varied significantly across patients. This confirmed that a intensity normalization step was a critical requirement for the preprocessing pipeline to prevent model bias.

5.3 Preprocessing Pipeline

A fully automated preprocessing pipeline was developed using the **MONAI** framework. This pipeline is a critical component of the project's reproducibility and robustness. For every patient, the following transformations were applied:

1. **LoadImage** : Loads the four `.nii.gz` MRI files and the segmentation mask.
2. **ConvertToMultiChannelBasedOnBratsClasses** : Converts the single-channel ground truth mask (with labels 1 for necrotic core, 2 for edema, 4 for enhancing tumor) into a three-channel binary mask for the three target labels: Whole Tumor (WT), Tumor Core (TC), and Enhancing Tumor (ET).
3. **Spacing** : Resamples all 3D volumes to a uniform isotropic voxel spacing of (1.0, 1.0, 1.0) mm. This ensures that the physical size of anatomical features is consistent across all scans.
4. **ScaleIntensityRange** : Normalizes the voxel intensity values of the MRI scans to a standard floating-point range of [0.0, 1.0], a crucial step for stable neural network training.
5. **CropForeground** : Automatically crops the image to remove large areas of empty background/air, focusing computational resources on the brain tissue.
6. **Resized** : Resizes all cropped volumes to a fixed spatial dimension of (128, 128, 128) . This creates uniform input tensors required by the deep learning model.

5.4 Multi-Modal Data Fusion

The project employed an **early fusion** strategy. After preprocessing, the four MRI modalities (T1, T1ce, T2, FLAIR) were stacked along the channel dimension, creating a single 4-channel, 3D tensor of shape `(4, 128, 128, 128)`. This tensor served as the direct input to the model.

5.5 Data Splitting

The preprocessed dataset was split into an 80% training set and a 20% validation set. The split was performed with a fixed random state (`RANDOM_STATE = 42`) to ensure that all experiments were conducted on the exact same data splits, guaranteeing reproducibility.

For testing, we are using unseen unique 150 samples from BRATS dataset which are not included in the 80/20 split.

6. Model Development and Hyperparameter Tuning

6.1 Model Architecture: 3D Attention U-Net

The final model chosen for this project was a **3D Attention U-Net**, implemented using MONAI. This choice was justified by:

- **Volumetric Processing:** Its 3D nature allows it to leverage inter-slice context, which is essential for anatomically coherent segmentations of 3D tumors.
- **Superior Localization:** The U-Net backbone with skip connections effectively combines deep semantic features with shallow spatial details, which is critical for precise boundary delineation.
- **Targeted Feature Selection:** The integrated attention gates force the model to learn to focus on the most relevant features for tumor segmentation, reducing false positives and improving performance.

The specific architecture was configured with 5 downsampling/upsampling levels, with channel dimensions of `(16, 32, 64, 128, 256)`.

6.2 Training Setup

All training was conducted in a Kaggle environment on an NVIDIA Tesla P100 GPU.

- **Loss Function:** A composite `DiceBCELoss` was used, calculated as `0.5 * DiceLoss + 0.5 * BinaryCrossEntropyWithLogitsLoss`. This combination balances the region-based optimization of Dice Loss (effective for class imbalance) with the pixel-wise stability of BCE Loss.
- **Optimizer:** The **AdamW** optimizer, a variant of Adam with improved weight decay regularization, was used.
- **Learning Rate Schedule:** An initial learning rate of `1e-4` was used with a polynomial decay scheduler.
- **Training Parameters:**
 - **Batch Size:** 2
 - **Total Number of Epochs:** 93 (determined by early stopping)
 - **Early Stopping:** Training was configured to halt if the validation Mean Dice Score did not improve for 7 consecutive epochs.
 - **Mixed Precision:** Automatic Mixed Precision (AMP) was enabled to accelerate training and reduce GPU memory consumption.

6.3 Hyperparameter Tuning and Experiments

A systematic approach was taken to arrive at the final training configuration:

- **Initial Experiments:** We explored K-Fold cross-validation and a 3D patch-based training approach. While K-Fold yielded slightly better but statistically similar results, its 5x computational cost was deemed too high. The patch-based approach suffered from being dominated by background accuracy, masking poor performance on the actual tumor sub-regions.
- **Hyperparameter Sweep:** A sweep was performed on the standard 80/20 split to optimize key parameters. The best configuration was found to be:
 - **Learning Rate:** 0.0001
 - **Batch Size:** 2
 - **Channels:** (16, 32, 64, 128, 256)
 - **Loss Weights:** (Dice: 0.5, BCE: 0.5)

6.4 Regularization and Optimization

To prevent overfitting and improve generalization, several techniques were employed:

- **Data Augmentation:** On-the-fly data augmentation was applied to the training set using MONAI's Rand transforms, including random flips, rotations, and intensity scaling.
- **Weight Decay:** The AdamW optimizer provided more effective weight decay than standard Adam.
- **Normalization:** ScaleIntensityRanged normalization was applied to all input scans.

7. Evaluation & Analysis

The final model, trained for 93 epochs, was subjected to a rigorous evaluation on an unseen test set of 150 patient scans.

7.1 Quantitative Results

The primary evaluation metric was the Dice Similarity Coefficient (DSC).

Label	Mean DSC	Std. Dev.	Median DSC	Min DSC	Max DSC
Tumor Core (TC)	0.5947	0.3489	0.7515	0.0000	0.9537
Whole Tumor (WT)	0.6892	0.3140	0.8393	0.0000	0.9544
Enhancing Tumor (ET)	0.5079	0.3267	0.6373	0.0000	1.0000
Mean Dice per Patient	0.5973	0.3057	0.7183	0.0000	0.9243

Key Insights from Quantitative Analysis:

- The model performs best on the **Whole Tumor (WT)**, achieving the highest mean and median Dice scores. This is expected as WT is the largest and most visually distinct target.
- The **Enhancing Tumor (ET)** is the most challenging region, with the lowest mean Dice score and the highest variance. The model completely failed to detect the ET region (Dice = 0) in **22 of the 150 test cases**, highlighting the difficulty of segmenting small, sparse, or low-contrast targets.
- **Volumetric Analysis:** While the model's predicted volumes showed a strong positive correlation with the ground truth volumes (Pearson correlation > 0.8), linear regression revealed a systematic

bias towards **under-predicting** the absolute tumor volume across all sub-regions.

7.2 Qualitative Analysis

Visual inspection of the segmentation results confirmed the quantitative findings.

- **Successful Cases:** In cases where the model achieved high Dice scores (>0.9), the predicted masks were visually almost indistinguishable from the ground truth. These typically involved patients with large, well-defined, and clearly enhancing tumors.
- **Failure Cases:** In cases with low Dice scores, the model either produced fragmented, noisy predictions or completely missed entire sub-regions (especially the ET). These failures were more common in patients with small tumors, diffuse boundaries, or atypical tumor appearances.

7.3 Error Analysis and Root-Cause Hypotheses

- **Systematic Bias:** The consistent under-prediction of volume suggests a potential calibration issue, possibly stemming from the loss function or post-processing thresholds.
 - **Class Imbalance:** The poor performance on the Enhancing Tumor is a classic symptom of class imbalance. The ET region often constitutes a very small percentage of the total brain volume, leading the model to be biased towards the more prevalent classes (background and non-enhancing regions).
 - **Architectural Limitations:** The model's receptive field or capacity may be insufficient to consistently capture the fine-grained details of very small or irregularly shaped enhancing tumor regions.
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8. Deployment & Documentation

8.1 System Architecture and Deployment

The final, trained model was integrated into a fully functional, end-to-end web application.

- **Backend:** A **Flask** server written in Python handles all core logic, including:
 - File uploads.
 - Execution of the MONAI preprocessing pipeline.
 - Inference with the loaded PyTorch model.
 - Post-processing of the prediction mask.
 - Generation of 2D slice images and 3D mesh data via API endpoints.
- **Frontend:** A single-page application built with **HTML, CSS, and JavaScript** provides the user interface. It communicates with the Flask backend via asynchronous API calls.
- **Deployment:** The application is containerized using Docker and is deployed on **Hugging Face Spaces**, making it publicly accessible for demonstration purposes. It can also be run locally with minimal setup.

8.2 User Workflow

The application provides a seamless workflow for tumor segmentation:

1. **Load Data:** The user can either upload their own set of four NIfTI files (FLAIR, T1, T1ce, T2) or select a pre-loaded example case.
2. **Run Prediction:** With a single click, the user can trigger the entire inference pipeline on the backend.

3. **Visualize Results:** The system presents the results in two interactive viewers:
- A **2D Slice Viewer** to inspect the input modalities and predicted masks on a slice-by-slice basis.
 - A **3D Viewer**, powered by Three.js, which renders the brain and the predicted tumor sub-regions as interactive 3D meshes.

8.3 Documentation

Comprehensive documentation was created to ensure the project is understandable, reproducible, and maintainable. This includes:

- **A Technical Document** detailing the environment setup, data pipeline, model architecture, and inference process.
 - **A User Guide** with step-by-step instructions for using the web application.
 - **Well-commented Code and README files** for both the training pipeline and the inference application.
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9. Conclusion and Future Work

9.1 Conclusion

Project Orca successfully achieved its primary objective of creating an end-to-end deep learning system for automated brain tumor segmentation. The 3D Attention U-Net model demonstrates promising performance, particularly in delineating the whole tumor and tumor core. The project's main contribution is the creation of a complete, deployable tool that bridges the gap from a raw MRI scan to an interactive, quantitative analysis, showcasing the potential of deep learning to enhance clinical workflows. While the model shows strong capabilities, the evaluation also highlighted the persistent challenges in medical imaging, particularly the accurate segmentation of small and variable sub-regions like the enhancing tumor.

9.2 Future Work

Based on the project's findings, several avenues for future improvement are proposed:

- **Advanced Model Architectures:** Explore an **ensemble model**, combining the predictions of the current Attention U-Net with a complementary architecture like a ResU-Net or a Vision Transformer, which could improve robustness and accuracy.
 - **Targeted Loss Functions:** Implement more advanced loss functions, such as the Focal Tversky Loss or boundary-aware losses, to specifically address the class imbalance even better and improve the segmentation of the Enhancing Tumor.
 - **Improved Post-processing:** Develop more sophisticated post-processing techniques, potentially incorporating anatomical priors to remove false positives and ensure that predictions are anatomically plausible.
 - **Generalization to Real-World Data:** To bridge the gap to clinical applicability, the model should be fine-tuned and validated on datasets from different institutions and MRI scanners to improve its generalization capabilities.
 - **Integration of GenAI for Reporting:** Fulfill the original secondary objective by integrating a Generative AI model (e.g., a fine-tuned LLM) to translate the quantitative volumetric outputs into a preliminary, human-readable clinical report summary.
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10. References and Appendix

10.1 References

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10.2 Project Team

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